

Prueba TestBench aplicada para verificar el funcionamiento de los filtros

Las pruebas aplicadas en este test son numéricas, ya que en el momento en que sea realizaron no se tenía audio real disponible. Este test verifica el comportamiento matemático que se espera para cada filtro.

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# === PRUEBA 1: Bypass â esperado out == in ===
# filter_sel=00 | sample_in= 1000 | sample_out= 1000 | valid=1
# [PASS] out=1000 == esperado=1000
# filter_sel=00 | sample_in= 2000 | sample_out= 2000 | valid=1
# [PASS] out=2000 == esperado=2000
# filter_sel=00 | sample_in= -1000 | sample_out= -1000 | valid=1
# [PASS] out=-1000 == esperado=-1000
# filter_sel=00 | sample_in= -2000 | sample_out= -2000 | valid=1
# [PASS] out=-2000 == esperado=-2000
# filter_sel=00 | sample_in= 0 | sample_out= 0 | valid=1
# [PASS] out=0 == esperado=0
#
# === PRUEBA 2: Lowpass escalon 0->8000 (debe subir hacia 8000) ===
# filter_sel=01 | sample_in= 8000 | sample_out= 1000 | valid=1
# [PASS] out=1000 == esperado=1000
# filter_sel=01 | sample_in= 8000 | sample_out= 1875 | valid=1
# [PASS] out=1875 == esperado=1875
# filter_sel=01 | sample_in= 8000 | sample_out= 2640 | valid=1
# [PASS] out=2640 == esperado=2640
# filter_sel=01 | sample_in= 8000 | sample_out= 3310 | valid=1
# [PASS] out=3310 == esperado=3310
# filter_sel=01 | sample_in= 8000 | sample_out= 3896 | valid=1
# [PASS] out=3896 == esperado=3896
# filter_sel=01 | sample_in= 8000 | sample_out= 4409 | valid=1
# [PASS] out=4409 == esperado=4409
# filter_sel=01 | sample_in= 8000 | sample_out= 4857 | valid=1
# [PASS] out=4857 == esperado=4857
# filter_sel=01 | sample_in= 8000 | sample_out= 5249 | valid=1
# [PASS] out=5249 == esperado=5249
# filter_sel=01 | sample_in= 8000 | sample_out= 5592 | valid=1
# [PASS] out=5592 == esperado=5592
# filter_sel=01 | sample_in= 8000 | sample_out= 5893 | valid=1
# [PASS] out=5893 == esperado=5893
#
# === PRUEBA 3: Highpass +-8000 (debe oscilar cerca de +-8000) ===
# filter_sel=10 | sample_in= 8000 | sample_out= 7000 | valid=1
# [PASS] out=7000 == esperado=7000
# filter_sel=10 | sample_in= -8000 | sample_out= -7875 | valid=1
# [PASS] out=-7875 == esperado=-7875
# filter_sel=10 | sample_in= 8000 | sample_out= 7110 | valid=1
# [PASS] out=7110 == esperado=7110
# filter_sel=10 | sample_in= -8000 | sample_out= -7778 | valid=1
# [PASS] out=-7778 == esperado=-7778
# filter_sel=10 | sample_in= 8000 | sample_out= 7195 | valid=1
# [PASS] out=7195 == esperado=7195
# filter_sel=10 | sample_in= -8000 | sample_out= -7704 | valid=1
# [PASS] out=-7704 == esperado=-7704
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# === PRUEBA 4: Bass Boost constante 4000 (debe superar 4000 y subir) ===
# DEBUG BASS | sample_in=4000 | low=0 | bass=4000 | selected=4000 | out=4000
#   filter_sel=11 | sample_in= 4000 | sample_out= 4250 | valid=1
#   [PASS] out=4250 == esperado=4250
# DEBUG BASS | sample_in=4000 | low=500 | bass=4250 | selected=4250 | out=4250
#   filter_sel=11 | sample_in= 4000 | sample_out= 4468 | valid=1
#   [PASS] out=4468 == esperado=4468
# DEBUG BASS | sample_in=4000 | low=937 | bass=4468 | selected=4468 | out=4468
#   filter_sel=11 | sample_in= 4000 | sample_out= 4659 | valid=1
#   [PASS] out=4659 == esperado=4659
# DEBUG BASS | sample_in=4000 | low=1319 | bass=4659 | selected=4659 | out=4659
#   filter_sel=11 | sample_in= 4000 | sample_out= 4827 | valid=1
#   [PASS] out=4827 == esperado=4827
# DEBUG BASS | sample_in=4000 | low=1654 | bass=4827 | selected=4827 | out=4827
#   filter_sel=11 | sample_in= 4000 | sample_out= 4973 | valid=1
#   [PASS] out=4973 == esperado=4973
# DEBUG BASS | sample_in=4000 | low=1947 | bass=4973 | selected=4973 | out=4973
#   filter_sel=11 | sample_in= 4000 | sample_out= 5101 | valid=1
#   [PASS] out=5101 == esperado=5101
#
# === PRUEBA 5: Saturacion positiva (32767) â warm-up 1, verif. 3 ===
# DEBUG BASS | sample_in=30000 | low=0 | bass=30000 | selected=30000 | out=30000
#   filter_sel=11 | sample_in= 30000 | sample_out= 31875 | valid=1
#   [WARM-UP 1] out=31875
# DEBUG BASS | sample_in=30000 | low=3750 | bass=31875 | selected=31875 | out=31875
#   filter_sel=11 | sample_in= 30000 | sample_out= 32767 | valid=1
#   [PASS] Saturacion correcta: 32767
# DEBUG BASS | sample_in=30000 | low=7031 | bass=33515 | selected=33515 | out=32767
#   filter_sel=11 | sample_in= 30000 | sample_out= 32767 | valid=1
#   [PASS] Saturacion correcta: 32767
# DEBUG BASS | sample_in=30000 | low=9902 | bass=34951 | selected=34951 | out=32767
#   filter_sel=11 | sample_in= 30000 | sample_out= 32767 | valid=1
#   [PASS] Saturacion correcta: 32767
#
# === PRUEBA 6: Saturacion negativa (-32768) â warm-up 1, verif. 3 ===
# DEBUG BASS | sample_in=-30000 | low=0 | bass=-30000 | selected=-30000 | out=-30000
#   filter_sel=11 | sample_in= -30000 | sample_out= -31875 | valid=1
#   [WARM-UP 1] out=-31875
# DEBUG BASS | sample_in=-30000 | low=-3750 | bass=-31875 | selected=-31875 | out=-31875
#   filter_sel=11 | sample_in= -30000 | sample_out= -32768 | valid=1
#   [PASS] Saturacion correcta: -32768
# DEBUG BASS | sample_in=-30000 | low=-7032 | bass=-33516 | selected=-33516 | out=-32768
#   filter_sel=11 | sample_in= -30000 | sample_out= -32768 | valid=1
#   [PASS] Saturacion correcta: -32768
# DEBUG BASS | sample_in=-30000 | low=-9903 | bass=-34952 | selected=-34952 | out=-32768
#   filter_sel=11 | sample_in= -30000 | sample_out= -32768 | valid=1
#   [PASS] Saturacion correcta: -32768
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# === PRUEBA 7: Lowpass escalon 0->-8000 (debe bajar hacia -8000) ===
# filter_sel=01 | sample_in= -8000 | sample_out= -1000 | valid=1
# [PASS] out=-1000 == esperado=-1000
# filter_sel=01 | sample_in= -8000 | sample_out= -1875 | valid=1
# [PASS] out=-1875 == esperado=-1875
# filter_sel=01 | sample_in= -8000 | sample_out= -2641 | valid=1
# [PASS] out=-2641 == esperado=-2641
# filter_sel=01 | sample_in= -8000 | sample_out= -3311 | valid=1
# [PASS] out=-3311 == esperado=-3311
# filter_sel=01 | sample_in= -8000 | sample_out= -3898 | valid=1
# [PASS] out=-3898 == esperado=-3898
# filter_sel=01 | sample_in= -8000 | sample_out= -4411 | valid=1
# [PASS] out=-4411 == esperado=-4411
# filter_sel=01 | sample_in= -8000 | sample_out= -4860 | valid=1
# [PASS] out=-4860 == esperado=-4860
# filter_sel=01 | sample_in= -8000 | sample_out= -5253 | valid=1
# [PASS] out=-5253 == esperado=-5253
#
# === PRUEBA 8: Cambio de filtro en caliente ===
# filter_sel=01 | sample_in= 4000 | sample_out= 500 | valid=1
# filter_sel=01 | sample_in= 4000 | sample_out= 937 | valid=1
# (cambiando a Bypass)
# filter_sel=00 | sample_in= 1234 | sample_out= 1234 | valid=1
# [PASS] out=1234 == esperado=1234
# filter_sel=00 | sample_in= -5678 | sample_out= -5678 | valid=1
# [PASS] out=-5678 == esperado=-5678
#
# =====
# RESUMEN: 43 PASS | 0 FAIL | 43 TOTAL
# =====
#

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